Bargaining over Leasing Contracts: Strong by Privilege but Weak by Risk Aversion

Kei Ikegami
New York University

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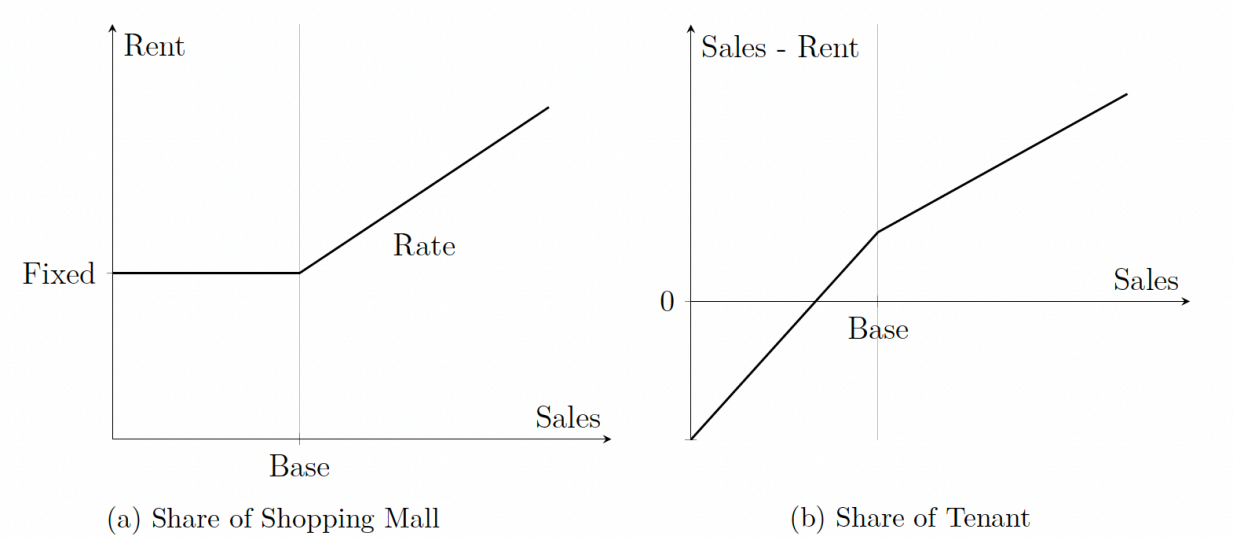
Fair Trading Environment

- **Regulatory shift**: focus on *fairness in individual transactions*, not just on preserving market-wide competition
  - Drivers: protecting SMEs and the rise of digital platforms
  - Examples: Germany's §20 GWB, Japan's JFTC Guidelines on Abuse of Superior Bargaining Position
- **Empirical challenge**: identifying unfair conduct in complex contracts
  - Overtly abusive clauses have largely disappeared, complicating detection
- **Example**: tenant leasing in shopping malls



(a) Share of Shopping Mall (b) Share of Tenant

Figure 1. How Sales is Shared with Shopping Mall and Tenant

Actual Cases in Tenant Leasing

- Many cases are found in U.S, EU, South Korea, Japan, etc...
- In particular, **Singapore**'s case is about the contract form itself:
  - A Fair Tenancy Industry Code of Conduct
    - Standards for balanced lease terms
  - It prevents landlords from
    - using multiple rent formulas
    - imposing one-sided termination rights

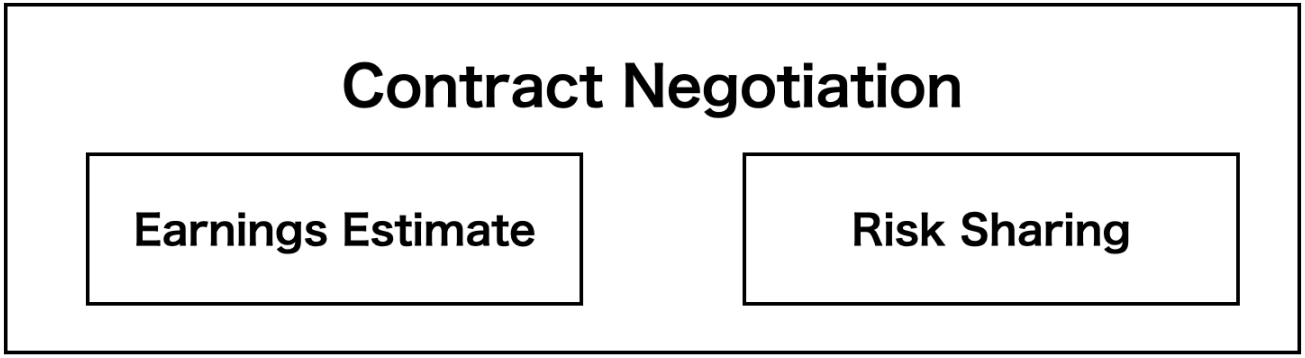
This Study

- A model of bargaining over leasing contracts in renewal negotiations
  - Two-stage sequential bargaining between the mall and the tenant
    - "Bargaining over earnings estimates" and "Bargaining over risk sharing"
  - Fundamental conflict:
    - The tenant seeks to justify a higher earnings estimate
    - The mall relies on past sales data and prefers to adhere to it
- The model is applied to real-world contract and sales data:
- Two opposing effects of the mall's superior position on contract form
  - A superior mall prefers variable rents and forgoes a higher fixed rent
  - A privileged mall is more risk-averse and favors a higher fixed rent
- Simulation: fairer bargaining does not always lead to less rent
  - In any cases, the total rent does not necessarily decrease
  - A sharp rise in the variable component by risk attitude adjustments

Data

- Data covers two shopping malls over 6 years (2017-2023)
  - Managed by same company, by different managers
  - Located in different commercial region in Japan
  - One in downtown (*Mall 0*) $\rightarrow$ More *privileged* mall
  - The other in suburb (*Mall 1*) $\rightarrow$ Less *privileged* mall
- For them, I have the following two data sources, linked as a panel data
  1. performance data
  2. contract data
- They are used for actual leasing operation of the management company
- I focus on contract renewals not contracts with new entrants
  - In total, 226 tenants operate under 443 contracts

Background of Contract Negotiation

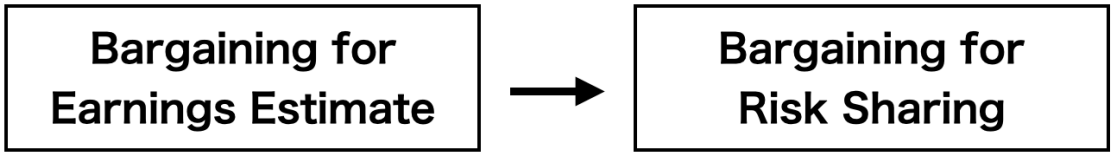


- Mall and Tenant has a negotiation table for a new contract term
  - including, rent structure, leasing period, and other things
  - but the main issue is rent structure
- Each contract negotiation is composed of two tasks
  - **Earnings estimate**: Derive the sales distribution for the next period
  - **Risk sharing**: Shift the risk involved in the estimate toward each other
- Earnings estimate is not an observed average sales: professionals say
  - Tenant tries to increase the earnings estimate
  - Mall sticks to the realized sales history

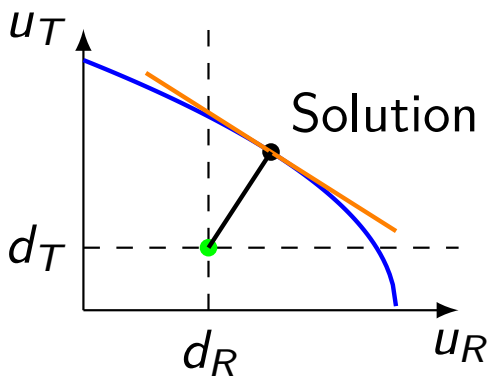
Motivating Observations and Anecdotes

- Risk attitude matters figure
  - "Risk sharing is the primary reason to set a share contract."
  - Clearly trading off the risk (variance of rent) against the volume of rent
  - I consider both tenant and mall is risk averse with different risk aversion
- Little concern for moral hazard
  - "It's hard to imagine that sales efforts would change based on contract"
    - **Tenant side**: Shop manager is evaluated by sales not rent
    - **Mall side**: Sales-enhancing efforts (e.g., cleaning) are funded separately through charges collected from tenants
  - No bunching at Base where the commission rate starts
  - No statistically significant effect of any contract terms on realized sales table

Model of Contract Negotiation



- The two tasks are sequentially resolved through separate Nash bargaining
- **Nash bargaining solution**
  - Primitives: Bargaining frontier, Bargaining power ratio, Break-up point
  - Outcome: Surplus split between parties



- Solution concept: **Nash-in-Nash solution**
  - Multiple interrelated Nash bargaining problems are solved under "other problems fall into the Nash bargaining solution" assumption
- Two BP ratios for the first table and the second table
  - Inverse of second BP ratio = Ratio of Risk Aversions
  - First BP ratio = *Privilege* ← Directly parametrized by covariates

Empirical Strategy: Variables

- An ongoing contract is denoted by a pair of i (tenant) and k (mall)
- Any contract is numerated with the contract number τ starting from 1
- Contract period (in months) is denoted by $T_{ik\tau}$
- Performance is measured for every months: $\tilde{X}_{ikt}$ for $t \in \{1, \cdots, T_{ik\tau}\}$
  - Main measure: *rank of sales per area within a mall*
- For an ongoing contract, we make a measure of performance:

$$X_{ik\tau} = \frac{1}{T} \sum_{t=1}^{T_{ik\tau}} \tilde{X}_{ikt}$$

  - the average of the performance measure during the contract period

Empirical Strategy: Parametrization / Estimation

- Logarithm of bargaining power ratio

$$\ln B_{ik\tau}^{EE} = X'_{ik\tau}\gamma + \varepsilon_{ik\tau}^{EE},$$

  - Three variables related with tenant; such as *rank of sales per area*
  - Four variables related with the shopping mall; such as *mall total sales*
  - $\varepsilon_{ik\tau}^{EE}$ is unobserved factor

- (Omit) Value of outside option = linear part + error term $\tilde{\varepsilon}^o$

- Two random terms are i.i.d. joint Normal: $(\epsilon^{EE}, \tilde{\varepsilon}^o) \sim N(0, \Sigma)$

- Likelihood function is constructed as in Tobit model:

$$\text{Likelihood} = \begin{cases} Pr(\text{observe contract } \theta_{ik\tau}) & \text{if continuation} \\ Pr(\text{exit}) & \text{otherwise} \end{cases}$$

- Maximize log-likelihood under the constraint of Nash bargaining solution
  - Earnings estimate is endogeneous

Result: More Privilege → Risk Averse → Favor Fixed Rent

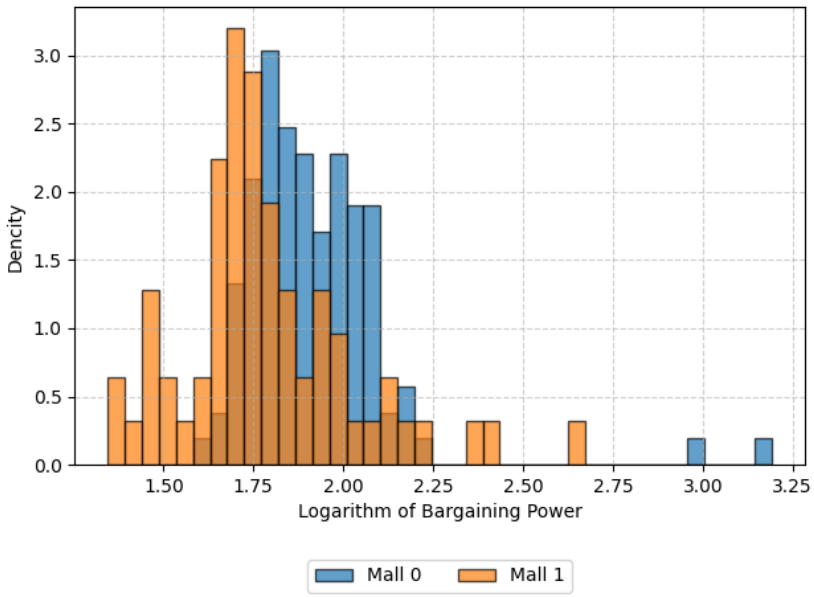


Figure: Privilege

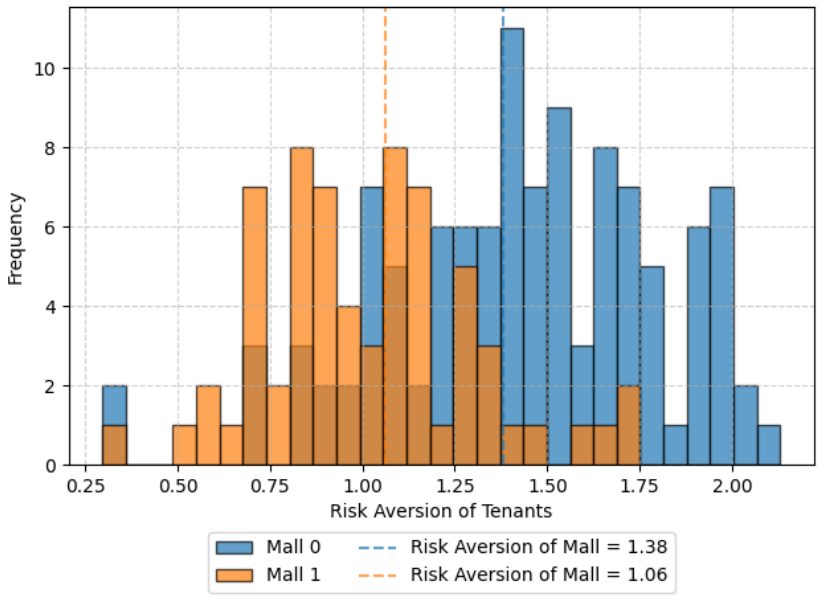


Figure: Risk aversion

- Mall 0 is more privileged due to high-traffic area
  → Earnings estimate is set to lower value
  → Commission component does not yield much rent
  → Mall manager emphasizes Fixed, i.e., she becomes more risk averse

Result: More Privilege → Forego Fixed Rent

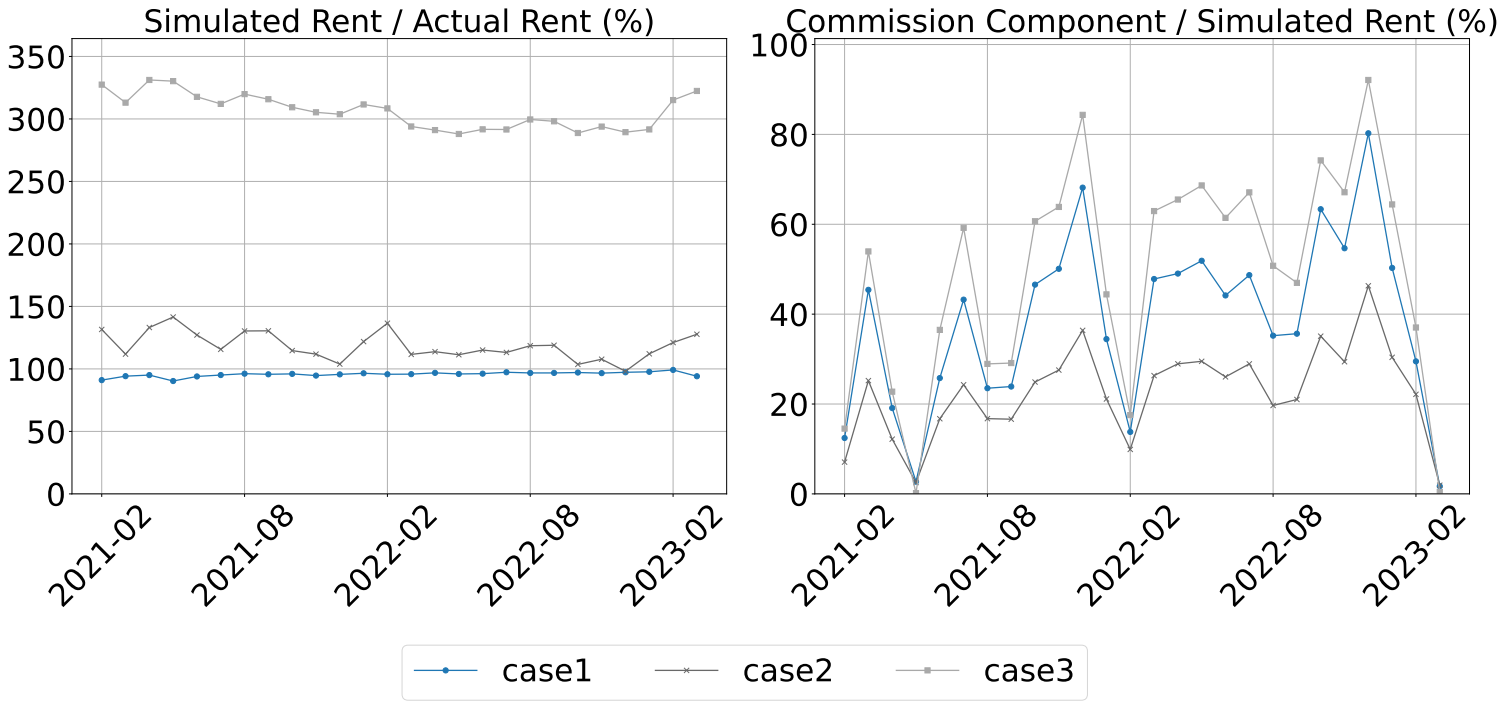
Mall	Mall 0			Mall 1		
Dep. Var	Rate	Base	Fixed	Rate	Base	Fixed
Avg. Sales	0.00204***	-0.135	0.00922	0.00533***	0.461***	0.0274
	(0.000720)	(0.0843)	(0.0134)	(0.000962)	(0.121)	(0.0202)
Var. Sales	0.000881	0.671***	0.0552**	-0.00307	-0.130	0.0299
	(0.00116)	(0.136)	(0.0216)	(0.00321)	(0.404)	(0.0675)
Privilege	-0.0480***	-2.987*	-0.0569	-0.0400***	-3.086**	-0.874***
	(0.0136)	(1.590)	(0.253)	(0.0101)	(1.275)	(0.213)
Risk loving	0.125***	-2.154	0.341	0.0985***	-1.350	-0.157
	(0.0121)	(1.416)	(0.225)	(0.00789)	(0.994)	(0.166)
N	90	90	90	62	62	62

- Privilege leads to smaller fixed and lower base
  - Why? Privileged mall is more pessimistic about earnings estimate
- Contract selection rule itself is consistent with risk attitude
  - Mall 0 (Risk averse): Large volatility → Higher Fixed
  - Mall 1 (Risk loving): Higher sales → Higher base and Higher rate

Simulation Setting: Fairer Trade

- Question: How will the transfer change in the fairer situation?
  - Why is this an empirical question? $\rightarrow$ Two paths exist
    1. Weaker positions $\rightarrow$ Higher earnings estimate $\rightarrow$ Larger fixed rent
    2. Weaker position $\rightarrow$ Less risk averse $\rightarrow$ More commission component
- Situation:
  - Mall 0 potentially abuses its privilege when making contracts
  - Regulator enforces the fairer bargainings in renewal through warnings
- Scenarios:
  - Case 1: Replicate actual rents
  - Case 2: Mall 0's privilege is determined in the same way as in Mall 1
  - Case 3: Case 2 + Mall 0's risk aversion is set to the same value of Mall 1

Result: Fairer bargaining does not always lead to less rent



- Case 2: weaker mall with fixed risk attitude
  - Drop in commission component; consistent with second mechanism
  - Increase in total rent: actual base may be too high
- Case 3: weaker mall becomes risk-loving
  - Rise in commission component: first mechanism dominates
  - Large increase in total rent revenue

Conclusion

- I analyze tenant contracts in shopping malls
- I find that fair bargaining has a complex influence on contract terms
  - A weaker-positioned mall may adopt riskier contract terms
- Regulators must be cautious when intervening in contract negotiations
  - Such interventions may lead to unexpected changes in contract terms
  - ... and the following transfers among them
- Future works:
  - Anchor tenant issue, i.e. not a bilateral bargaining situation
  - Moral hazard issue

Appendix

Obs. : Risk Attitude Matters

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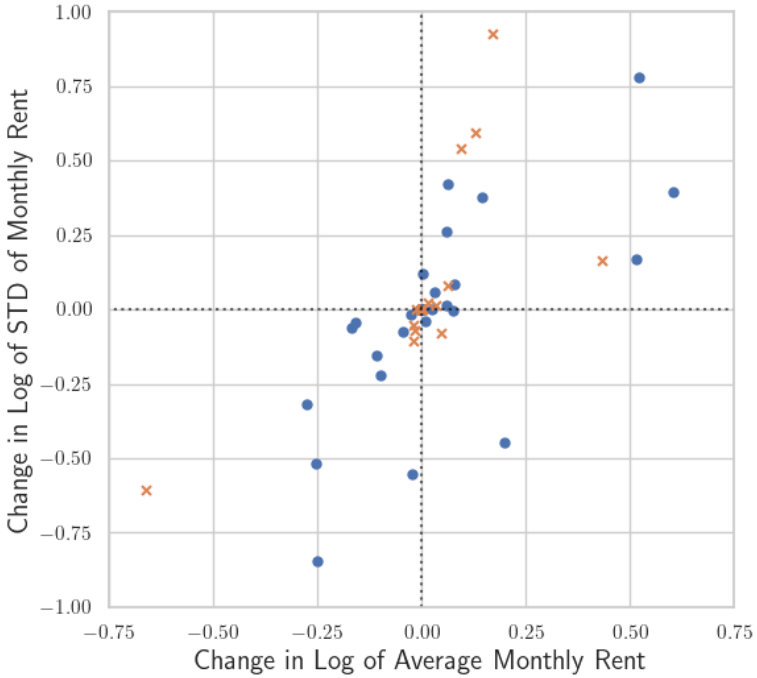
- For all contracts:
  - Average monthly rent
  - Standard deviation of monthly rents
- Changes in these metrics over two successive contracts are plotted
- Higher rent must be accompanied by larger variance



Moral Hazard?

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Dep. Var. : Sales	(1) Level	(2) Diff	(3) Diff / Change
Rate	−0.063	−0.049	−0.180
	(0.219)	(0.222)	(0.374)
Base	0.069	0.000	0.000
	(0.206)	(0.000)	(0.000)
Fixed	2.839**	0.000	0.000
	(1.319)	(0.000)	(0.000)
Observations	197	197	156
Adjusted R^2	0.992	0.964	0.967

- Moral hazard test: Do contract terms affect tenant performance?
  - Regress average monthly sales on contract terms and controls
  - No evidence that higher commission rates reduce tenant effort
- Institutional features mitigate moral hazard: mystery shopper surveys assess

Identification

- Two sets of parameters
  1. Common across tenants within a mall
     - risk aversion of mall and marginal effects of covariates etc
  2. Contract-specific
     - Risk aversion of tenant: $\rho_{i\tau}$
     - Agreed upon expected sales (endogeneous parameter): $\mu_{ik\tau}$
- The first set is identified by the variation of exit offer
  - The second group is removed from the continuation decision
  - WHY?
    - Assumption $B^{RS} = \frac{\rho_T}{\rho_R}$ eliminates ρ_R from the decision
    - Cooperative approach allows me to parametrize $B^{EE}_{ik\tau}$ determining $\mu_{ik\tau}$
- The second set is identified by conditions of Nash solution
  - Two agents $\rightarrow$ Two surplus expressions $\rightarrow$ Two conditions